Midterm Exam — Practice Form

ECON 470 — Microeconomics of Artificial Intelligence

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Instructions.

- Time: **110 minutes**. Closed book, closed notes; a calculator is allowed. No phones or laptops.
- Total points: **100**. Point values are listed next to each question. Allocate your time accordingly.
- Show your work for full credit. Partial credit is given for correct setup even when arithmetic is off.
- Where a question asks for a number, leave fractions or decimals to three significant figures. Natural logs and exponentials may be left unsimplified (e.g., $-5\ln(0.3)$) if a calculator is not handy.
- If you make a simplifying assumption to keep moving, state it clearly.

Problem 1: Concepts [15 pts]

Answer each part in **2–3 sentences**. Plain prose, no equations needed unless you want to use one.

(a) *[3 pts]* In Lab 1 you trained microGPT on dinosaur names and you trained microOLS to estimate a demand elasticity for orange juice. Both used the *same* `Value` class, the same backpropagation, the same Adam optimizer. In what sense is each program doing "prediction"? What is the analogue of "the next character" in the orange-juice regression?

(b) *[3 pts]* A symmetric signal s with accuracy e is called **informative** when $e > 1/2$. Explain in plain language what fails when $e = 1/2$ and why such a signal cannot change a decision-maker's optimal action.

(c) *[3 pts]* The value of a (symmetric) signal in the 2×2 model is $V(e) = (e - p)(R - r)$. State two distinct comparative statics this formula implies (e.g., what happens to V as e rises, as $R - r$ widens, etc.) and give a one-sentence economic interpretation of each.

(d) *[3 pts]* In Lab 3 you found that a firm with optimal protection x^*_{RULE} adopts AI only when $e > p + x^*_{RULE}$, **not** the looser condition $e > p$. Explain in plain language *why* protection raises the bar for AI adoption.

(e) *[3 pts]* In Lab 4 (Gans Ch. 5) you saw that prediction alone has zero stand-alone value at the running parameters: $V_P - V_\emptyset = 0$. Yet prediction *combined* with judgment is highly valuable. In one or two sentences, explain why this is so and how that gives credence to the view that prediction makes experimental results "actionable."

Problem 2: A 2$×$2 Prediction Problem [30 pts]

A pediatrician sees a child who may or may not have strep throat. Let $\theta = 1$ denote "strep present" and $\theta = 2$ denote "strep absent." The prior is $p = \Pr(\theta = 1) = 0.6$.

The pediatrician must choose to **prescribe antibiotics** ($a = 1$) or **not prescribe** ($a = 2$). The payoff is $R = 80$ when the action matches the state and $r = 20$ otherwise.

A rapid antigen test produces a signal $s \in \{1, 2\}$ with symmetric accuracy $e_1 = e_2 = e = 0.8$.

Part A: Posteriors and informativeness [9 pts]

(a) *[3 pts]* Write out the four joint probabilities $\Pr(\theta = i, s = j)$ in a 2×2 table. Verify they sum to 1.

(b) *[3 pts]* Compute $\Pr(\theta = 1 \mid s = 1)$ and $\Pr(\theta = 1 \mid s = 2)$.

(c) *[3 pts]* Verify that the test is informative by comparing each posterior to the prior $p = 0.6$. In one sentence, explain what "informative" means in terms of beliefs.

Part B: Strategies and the value of the test [12 pts]

(d) *[3 pts]* Compute the expected payoff under the **best rule** (the rule that ignores the signal and always takes the better action).

(e) *[6 pts]* Compute the expected payoff under the strategy "follow the signal" ($a = s$). Show your work cell-by-cell.

(f) *[3 pts]* Compute the **value of the test**, $V(e) = E[U_{\text{follow}}] - E[U_{\text{rule}}]$. Verify against the formula $V(e) = (e - p)(R - r)$.

Part C: Demand for the test [5 pts]

A clinic has 500 pediatricians, all facing this same problem with $p = 0.6$ and $e = 0.8$. They differ only in stakes: $R_i - r_i$ is distributed uniformly on $[20, 120]$ across them.

(g) *[3 pts]* Write the willingness-to-pay (WTP) of pediatrician i as a function of $R_i - r_i$. At what price P does *no one* adopt? At what price does *everyone* adopt?

(h) *[2 pts]* At a price $P = 12$, how many of the 500 pediatricians adopt? (Show the cutoff and the count.)

Part D: Reading an ROC curve [4 pts]

(i) *[4 pts]* The diagonal line TPR = FPR in the ROC diagram corresponds to $e_1 + e_2 = 1$. (i) In one sentence, explain why a signal *on* this diagonal is useless. (ii) In one sentence, explain what it means geometrically for "AI to get better" in this diagram.

Problem 3: Substitutes for Prediction [30 pts]

A firm faces a 2×2 environment with $R = 100$, $r = 20$, and prior $p = 0.6$.

The firm may invest in two forms of risk management:

- **Protection** $x \geq 0$: shifts $\Pr(\theta = 1)$ from p to $p + x$, at cost

$$C(x) = 1 \cdot \left[\frac{1}{1-p-x} - \frac{1}{1-p}\right] = \frac{1}{0.4-x} - \frac{1}{0.4}, \qquad C'(x) = \frac{1}{(0.4-x)^2}.$$

- **Insurance** $\Delta \geq 0$: raises the mismatch payoff from r to $r + \Delta$, at cost

$$c(\Delta) = -5\ln\left(1 - \frac{\Delta}{80}\right), \qquad c'(\Delta) = \frac{5}{80-\Delta}.$$

So $A = 1$ and $B = 5$ in the notation of Lab 3.

Without prediction, the firm follows the **rule** $a = 1$ (matches the more likely state).

Part A: The rule with optimal risk management [12 pts]

(a) *[2 pts]* Write the firm's expected payoff $\pi(x, \Delta)$ under the rule $a = 1$.

(b) *[3 pts]* Take the FOCs with respect to x and Δ.

(c) *[4 pts]* Solve for x^*_{RULE} and Δ^*_{RULE}. (Hint: divide the two FOCs.)

(d) *[3 pts]* Compute the firm's expected payoff at $(x^*_{RULE}, \Delta^*_{RULE})$. You may leave ln unsimplified.

Part B: AI arrives [10 pts]

An AI prediction system becomes available with accuracy $e = 0.75$ and per-period cost $\lambda = 2$. Under a **decision** ($a = s$), the firm sets $x = 0$ (Proposition 4.1).

(e) *[3 pts]* Solve for Δ^*_{DEC} from the FOC $1 - e = B/(R - r - \Delta)$.

(f) *[3 pts]* Compute the firm's expected payoff under the decision: $eR + (1 - e)(r + \Delta^*_{DEC}) - c(\Delta^*_{DEC}) - \lambda$.

(g) *[2 pts]* Should the firm adopt AI? Compare the rule and decision payoffs.

(h) *[2 pts]* Check the condition $e > p + x^*_{RULE}$. Briefly: why is *this*, not $e > p$, the relevant adoption threshold?

Part C: Better AI [8 pts]

The firm now has access to a better signal with $e = 0.9$ at the same cost $\lambda = 2$.

(i) *[3 pts]* Solve for the new Δ^*_{DEC} and compute the new decision payoff.

(j) *[3 pts]* Should the firm adopt the better AI? By how much does the expected payoff change relative to the rule?

(k) *[2 pts]* Did insurance rise or fall when the firm switched from the rule to the decision in this case? Is this consistent with Proposition 4.1?

Problem 4: Complements to Prediction [25 pts]

An agent faces the two-state model from Gans Ch. 5 / the Lab 4 handout: states $\theta \in \{1, 2\}$, each equally likely; safe action paying S; risky action paying R in some states and r in others; common probability v that risky pays R in any given state. Running parameters for this problem:

$$R = 100, \quad r = 20, \quad S = 70, \quad v = 0.4, \quad \delta = 0.9.$$

Part A: The assumptions [6 pts]

(a) *[3 pts]* Verify Assumption A1 (Safe Default): $vR + (1 - v)r \leq S$. In one sentence, state what A1 rules out and why the complementarity story would collapse without it.

(b) *[3 pts]* Verify Assumption A2 (Judgment Insufficient): $\frac{1}{2}(R + r) \leq S$. In one sentence, state what A2 means in plain language.

Part B: The four cells [9 pts]

Use the formulas $V_\emptyset = V_P = S$, $V_J = v^2R + (1-v^2)S$, $V_{PJ} = vR + (1-v)S$.

(c) *[3 pts]* Compute $V_\emptyset$, V_P, V_J, V_{PJ} at the running parameters.

(d) *[3 pts]* Compute the stand-alone value of prediction $V_P - V_\emptyset$ and the stand-alone value of judgment $V_J - V_\emptyset$. Compute the *combined* value $V_{PJ} - V_\emptyset$. What share of the combined value is *complementarity* (the surplus above the sum of the two stand-alone values)?

(e) *[3 pts]* In 1–2 sentences, explain in plain language why $V_P - V_\emptyset = 0$ here, even though $V_{PJ} - V_J > 0$.

Part C: The threshold [10 pts]

The threshold for the agent to find delayed, uncertain experimentation worthwhile *given prediction* is

$$\bar{\lambda} = \frac{S(1-\delta)}{\delta v(R-S)}.$$

(f) *[3 pts]* Compute $\bar{\lambda}$ at the running parameters.

(g) *[2 pts]* Now suppose patience rises to $\delta = 0.99$ (everything else fixed). Recompute $\bar{\lambda}$. Does it rise or fall? One-sentence interpretation.

(h) *[2 pts]* Suppose instead the upside widens, $R = 130$ (so $R-S = 60$), with $\delta = 0.9$. Recompute $\bar{\lambda}$. Who has a stronger incentive to invest in prediction — a firm with a large or small $R-S$?

(i) *[3 pts]* Without prediction, the threshold is $\bar{\lambda}_{\text{no pred}} = \frac{S(1-\delta)}{\delta v^2(R-S)}$. Compute it at the *original* running parameters. In one sentence, explain what your number tells you about whether the agent would experiment in the absence of prediction.

End of practice exam.